

FedIn-NID: A Federated Learning Framework for Network Intrusion Detection in Large-Scale Heterogeneous Industrial IoT

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Abstract—The evolving Industrial Internet of Things (IIoT) is shifting towards decentralized collaborative manufacturing, posing heightened network security issues within interconnected value chains, thus requiring advanced Network Intrusion Detection (NID) systems to identify potential threats. In this context, traditional centralized NID systems are insufficient due to cross-industrial privacy concerns and interconnected secure threats. Federated Learning (FL) has emerged as a promising solution to enable the sharing of security insights without compromising privacy across participants. However, establishing an FL-based NID framework in realistic IIoT scenarios faces several hurdles, including the limited availability of large-scale devices and heterogeneous attack data distributions. The former leads to inconsistent client participation and degraded performance, while the latter hinders model convergence. To address these, we propose a novel Federated Learning-based Industrial Network Intrusion Detection (FedIn-NID) framework, incorporating a multidimensional client selection strategy and a dynamic global aggregation strategy. The selection strategy synergistically considers multidimensional factors including client availability, local dataset distribution, and dataset size. This approach accommodates clients with varying availability and avoids the selection of biased clients with data concentrated in a few categories. During model aggregation, the proposed strategy leverages the concept of exponential moving average to dynamically balance the holistic yet slightly older knowledge in the global model with the partial but relatively newer knowledge in local models, ensuring effective aggregation and convergence of the global NID model. Experiments demonstrate that FedIn-NID outperforms baselines by 10% to 30%, showcasing remarkable robustness with increasing data distribution heterogeneity and device count.

Index Terms—Federated learning, industrial Internet of Things (IIoT), network intrusion detection.

I. INTRODUCTION

THE advent of Industry 4.0 marks a transformative era in both manufacturing and service sectors, with the Industrial Internet of Things (IIoT) serving as its cornerstone [1]. Enabled by advanced wireless technologies like 5G and Beyond 5G (B5G), the IIoT landscape is increasingly characterized by cross-industrial collaborative manufacturing, allowing diverse entities and stakeholders to join forces for resource sharing and cooperative work [2]. While this integrated approach offers enormous benefits in terms of operational efficiency and innovation, it introduces unique network security challenges. In a collaborative manufacturing ecosystem, entities not only share resources but also integrate their data streams and control systems, increasing the surface area for potential cyber-attacks, thereby exposing their collaborators to new and diverse threat landscapes. Subsequently, to efficiently identify and counteract the various types of security threats in the foreseeable collaborative IIoT paradigm, the critical role of Network Intrusion Detection (NID) has been magnified [3].

NID is a security mechanism designed to monitor and analyze network traffic for signs of malicious attacks or security policy violations. However, traditional NID methodologies, such as signature-based and rule-based systems [4], [5], are becoming increasingly inadequate and less scalable in collaborative manufacturing facilitated by the complex interconnection of systems and attack-type imbalances. With the development of powerful deep learning techniques, specifically exemplified by Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), existing works have turned to learning-based NID systems [6], [7], [8]. These deep learning architectures are adept at modeling complex patterns and extracting hierarchical features from data, making them intrinsically suited for identifying a wide array of intrusion activities in the next-generation IIoT. In [9], Shone et al. introduced nonsymmetric deep autoencoder and random forest classification for NID. In [10], Liang et al. proposed an industrial NID algorithm based on multifeature data clustering optimization model. In [11], Lin et al. introduced an intrusion detection system featuring stacked sparse autoencoders and

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RNNs. In [12], He et al. proposed an intrusion detection model that leverages multi-modal deep autoencoders and LSTM to effectively integrate features from different levels of network connections. In [13], Hu et al. proposed a deep learning-based one-class intrusion detection scheme to improve the security of industrial networks. In [14], Andresini et al. introduced a method that represents network flows as images and obtains its imagery representation by combining nearest neighbor search and clustering processes. Thanks to the nonlinearity approximation capabilities of neural networks, the learning-based NID methodologies can effectively extract latent features and patterns from IIoT networks, and excels at detecting intrusions.

Nonetheless, these approaches necessitate extensive datasets for effective training among different entities, a requirement that often involves handling diverse data characteristics due to the geographical distribution variances of these industrial entities [15]. Further complicating matters is the sensitive nature of IIoT data, which demands adherence to rigorous privacy regulations and ethical guidelines, thereby obviating the feasibility of centralized data pooling for model training [16]. In light of these constraints, Federated Learning (FL) [17] emerges as a promising solution. FL aligns seamlessly with the distributed and edge-centric characteristics of contemporary IIoT systems. By retaining data locally on edge devices, FL not only fulfills privacy and regulatory mandates but also accommodates the inherent geographical distribution diversity, allowing for more contextually relevant model development without the need for centralized data aggregation. Additionally, FL facilitates the deployment of robust deep learning models across a distributed network without the necessity for centralized data collection, thus offering a holistic resolution to the multifaceted challenges of network security in distributed IIoT.

Since its inception, research in FL has rapidly evolved, with a significant focus on the security and robustness of the learning process itself. For instance, to defend against model poisoning attacks, Ma et al. [18] proposed ShieldFL, a defense using homomorphic encryption to identify malicious encrypted gradients, while Yazdinejad et al. [19] developed a model with an internal auditor using statistical methods for the same purpose. Other works focus on securing the aggregation process; Jebreel et al. [20] presented fragmented federated learning to reconcile accuracy with security by mixing update fragments, and Eltaras et al. [21] introduced an efficient protocol that is robust against user dropouts. From an attack perspective, research has also revealed new vulnerabilities, such as the External Gradient Inversion Attack (EGIA) proposed by Liang et al. [22], which shows that external adversaries can reconstruct private data from intercepted gradients.

While securing the FL process against such attacks is a critical long-term goal, the practical application of FL to a demanding domain like NID first requires overcoming more foundational challenges that can prevent the model from converging effectively. This has motivated a significant line of work focused on making FL viable for NID in the first place. For example, Mothukuri et al. [23] introduced an FL approach for intrusion detection in IoT networks, Liu et al. [24] combined FL with blockchain technology to develop an

intrusion detection system for securing vehicular networks, Sun et al. [25] proposed a hierarchical FL intrusion detection system with transformer-based local models, and Ruzafa et al. [26] evaluated the use of various differential privacy methods in FL-based network intrusion detection.

Despite these pioneering efforts, the application of FL-based network intrusion detection in realistic (i.e., large-scale and heterogeneous) IIoT environments still faces two main challenges that hinder model performance:

- 1) *Massive devices as clients with limited availability*¹: Though massive devices are interconnected in IIoT networks, not all devices can maintain consistent active states in each communication round due to resource constraints on access devices and communication overhead costs on the server. This results in inconsistent selection of clients over iterations, leading to divergent directions during the model aggregation process, potentially causing the trapped in the sub-optimal global model.
- 2) *Data heterogeneity*: The data on each device is inherently heterogeneous. This stems from intrinsic factors, such as different device types generating unique benign traffic patterns, as well as external factors like diverse attack distributions, where different devices are exposed to various types of attacks. This overall heterogeneity means the local data of any specific device does not represent the global data distribution.

Most existing methods [23], [24], [25] have not explicitly handled data heterogeneity or provided detailed insights into the scale and quantity of participating clients for FL in NID. While [26] addresses data heterogeneity, the application scenario is primarily confined to small-scale clients scenario (5 to 10 clients). There exists a pronounced research gap for an FL-based framework capable of concurrently addressing these challenges in the large-scale heterogeneous IIoT environment.

Motivated by this, in this paper, we propose an effective and robust **Federated Learning-based Industrial Network Intrusion Detection (FedIn-NID)** framework. We first propose a novel client selection strategy that synergistically integrates multi-dimensional factors (including client availability, local dataset distribution, and dataset size) to ensure efficient and balanced client participation. By leveraging these factors, the approach effectively evaluates the consistency of participating clients and avoids the selection of biased clients with data concentrated in a few categories. Then, considering the heterogeneity of attack data distributions, we introduce FedEMA, a dynamic aggregation strategy that leverages the concept of Exponential Moving Average (EMA) to dynamically update the global model. This innovative approach harmoniously integrates both previously learned global knowledge and newly acquired local knowledge, consequently accelerating the convergence of the global model and improving the performance of the detection model. Our contributions are summarized as follows:

- 1) *Comprehensive FedIn-NID Framework for IIoT Environments*: We propose a FedIn-NID framework to

¹Limited availability means that not all devices can maintain consistent active states in each communication round due to interruptions caused by changes in device state, system quotas, and network connectivity [15], [27].

address the comprehensive challenges in the realistic IIoT environment characterized by heterogeneous data distribution, inherent massive devices as clients, and limited availability across devices. Additionally, we contribute a 1D-CNN model for efficient network intrusion detection, enhancing the practicality and applicability of our framework in real industrial settings.

- 2) *Multidimensional Client Selection Strategy*: We introduce a client selection strategy that synergistically integrates multidimensional factors, including client availability, local dataset distribution, and dataset size. By leveraging a selection weight derived from these factors, the strategy effectively mitigates the consistency among participating clients and avoids the selection of biased clients with data concentrated in a few categories, ultimately enhancing the performance of FL in realistic and dynamic IIoT scenarios.
- 3) *Global Model Aggregation Strategy Using FedEMA*: We present a dynamic global model aggregation strategy, termed FedEMA. This strategy employs the concept of exponential moving average to dynamically balance the holistic yet slightly older knowledge in the global model with the partial yet relatively newer knowledge in local models. Consequently, FedEMA achieves a harmonious integration of knowledge acquired by the global model and local models, accelerating the convergence of the global model and improving the model performance.

Our experiments demonstrate that our proposed FedIn-NID achieves efficient detection of network intrusions amidst the large-scale heterogeneous IIoT environment. Specifically, FedIn-NID exhibits outstanding performance by 10% to 30% over baselines. Compared to other methods, our approach excels in detecting rare categories of attack instances, emphasizing its effectiveness in the heterogeneous data distribution environment. Furthermore, our method exhibits significant robustness in diverse heterogeneous data distribution settings and varying client numbers, confirming its superior adaptability to the complexity of realistic IIoT environments.

The remainder of the paper is organized as follows: Section II provides the system modeling. Section III describes the proposed FedIn-NID framework, including the proposed multidimensional client selection strategy, the designs of the model structure, local training, and the FedEMA strategy. Section IV presents the experimental setup and results. Finally, we conclude the paper with a summary in Section V.

II. SYSTEM MODEL

In this section, we illustrate the client model and network intrusion model in the large-scale heterogeneous IIoT scenario.

A. Client Model

The large-scale heterogeneous IIoT environment is illustrated on the right side of Fig. 1. We denote a total of M clients (devices) as $\mathcal{C} = \{C_1, \dots, C_M\}$. Each client C_i has a local dataset (collected data) $\mathcal{D}_i$ comprising N_i data pairs, defined as $\mathcal{D}_i = \{(\mathbf{x}_{i,j}, \mathbf{y}_{i,j})\}_{j=1}^{N_i}$, where $\mathbf{x}_{i,j} \in \mathcal{X} \subset \mathbb{R}^d$ represents a network flow with d features, and $\mathbf{y}_{i,j} \in \mathcal{Y} \subset \{0, 1, \dots, K-1\}$ denotes

the flow type, with the number of classes for network flows (benign and other attack types) denoted by K . The distribution of this data is naturally heterogeneous, influenced by both the intrinsic characteristics of the client device (e.g., its function and resulting benign traffic patterns) and its unique exposure to different types of network attacks. Following [28] and [29], we perform Dirichlet distribution for train to generate Non-IID data partitions. Specifically, consider the data distribution as follows:

$$\boldsymbol{\pi} = [\boldsymbol{\pi}_0, \boldsymbol{\pi}_1, \dots, \boldsymbol{\pi}_{K-1}]^T, \quad (1)$$

where $\boldsymbol{\pi}_k$ ($k \in \{0, 1, \dots, K-1\}$) is the data distribution for M clients at the k^{th} class as follows:

$$\boldsymbol{\pi}_k \sim \text{Dir}(\boldsymbol{\pi}_k | \boldsymbol{\gamma}_k) = \frac{1}{\mathbf{B}(\boldsymbol{\gamma}_k)} \prod_{j=1}^M \pi_{k,j}^{\gamma_{k,j}-1}, \quad (2)$$

where $\boldsymbol{\pi}_k = [\pi_{k,1}, \pi_{k,2}, \dots, \pi_{k,M}]$, and $\pi_{k,j}$ is the proportion of the instances of the k^{th} class for the j^{th} client; $\boldsymbol{\gamma}_k = [\gamma_{k,1}, \gamma_{k,2}, \dots, \gamma_{k,M}] > \mathbf{0}$ is the parameter of the Dirichlet distribution; $\mathbf{B}(\boldsymbol{\gamma}_k)$ is the multivariate beta function for normalization. Hence, $\boldsymbol{\pi} \sim \text{Dir}(\boldsymbol{\gamma})$ denotes the data distribution across M clients for K classes, where $\boldsymbol{\gamma} = [\boldsymbol{\gamma}_0, \boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_{K-1}]^T = \boldsymbol{\gamma} \cdot \mathbf{1}_{K \times M}$. For ease of presentation, we adopt $\text{Dir}(\boldsymbol{\gamma})$ to denote the Dirichlet distribution with the scalar hyper-parameter $\boldsymbol{\gamma}$.

Then, we follow [30], [31] to model the availability of clients in a heterogeneous IIoT environment. The availability of clients are affected by their current conditions, basically including their state of energy, communication costs for FL, and computational costs for local training. To facilitate comparison among those factors, we project them to the range of 0 to 1 without loss of generality.

1) *State of Energy* ($\mathcal{O}_{\text{energy},i}$): The state of energy can be estimated based on the current and maximum energy capacity of the client, as outlined below:

$$\mathcal{O}_{\text{energy},i} = e^{-\gamma \frac{E_i}{E_{\text{Max},i}}}, \quad (3)$$

where E_i is the remaining energy of client C_i , $E_{\text{Max},i}$ denotes its maximum energy capacity, and γ is the scaling factor.

2) *Communication Cost* ($\mathcal{O}_{\text{comm},i}$): In this paper, the communication cost is evaluated by the quality of transmission (i.e., transmission rate) and can be modeled as:

$$\mathcal{O}_{\text{comm},i} = \frac{r_{\text{Min}}}{r_i}, \quad (4)$$

where $r_i \in [r_{\text{Min}}, r_{\text{Max}}]$ is the transmission rate. A higher rate indicates lower communication cost for FL.

3) *Computation Cost* ($\mathcal{O}_{\text{comp},i}$): The computation cost is based on the client's available computational capability, such as CPU or GPU performance per training task. The higher the computational capability, the more efficiently the client can handle tasks, resulting in a lower cost. The cost can be computed as follows:

$$\mathcal{O}_{\text{comp},i} = \frac{P_{\text{Max}}}{P_i} \quad (5)$$

where $P_i \in [P_{\text{Min}}, P_{\text{Max}}]$ is the computational power. Note that $r_{\text{Min}}, r_{\text{Max}}, P_{\text{Min}}, P_{\text{Max}}$ are evaluated from the bound of total clients $\mathcal{C}$.

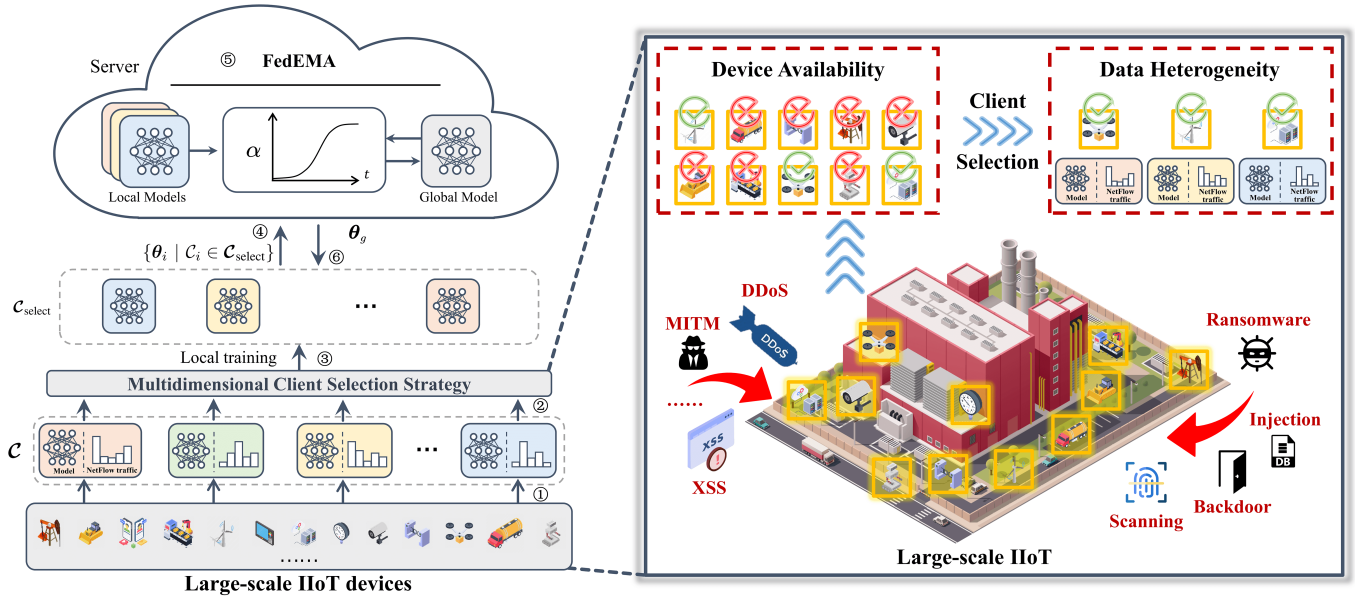


Fig. 1. An overview of our proposed FedIn-NID framework in the large-scale heterogeneous industrial IoT environment.

To summarize, the total cost for client C_i can be calculated as follows:

$$\mathcal{O}_i = \mathbf{w}^T \mathcal{O}, \quad (6)$$

where $\mathcal{O} = [\mathcal{O}_{\text{energy},i}, \mathcal{O}_{\text{comm},i}, \mathcal{O}_{\text{comp},i}]^T$ is the cost vector for client C_i , and $\mathbf{w} = [w_{\text{energy}}, w_{\text{comm}}, w_{\text{comp}}]^T$ is the corresponding weight vector that indicate the importance of each cost, and device availability for C_i is given by:

$$\mathcal{A}_i = \frac{1}{1 + \mathcal{O}_i} \quad (7)$$

where higher weighed costs lead to lower device availability.

B. Network Intrusion Model

Each client C_i has a local multi-class NetFlow traffic classification model θ_i , and the goal is to collaboratively train a good multi-class NetFlow traffic classification model θ_g by leveraging the isolated local datasets for efficient NID. NetFlow collects data about the IP traffic passing through it. This data is aggregated into what are called “flows”. A flow is typically defined by several key attributes, such as source/destination IP address and port, network protocol, type of service, etc. Based on NetFlow, the following attack types are considered in this paper besides “benign” flows:

- 1) Scanning: An unusually high number of flows indicating an attempt to discover accessible devices or services.
- 2) Cross-Site Scripting (XSS): Irregular flow patterns to web servers, indicating injection of malicious scripts.
- 3) Distributed Denial of Service (DDoS) & Denial of Service (DoS): A surge in flow volume, targeting specific destinations, overwhelming the network resources.
- 4) Password Attacks: Repeated flows to authentication services, suggesting brute-force attempts.
- 5) Injection: Unusual or unexpected flows to database servers, attempting to exploit database vulnerabilities.

- 6) Backdoor: Anomalous flows to uncommon ports or destinations, indicating traffic to unauthorized entry points.
- 7) Man-In-The-Middle (MITM): Anomalies in flow data that may suggest interception or alteration of traffic.
- 8) Ransomware: Abnormal flow patterns related to encryption or data exfiltration activities.

Given the pressing need to enhance NID accuracy while concerning clients’ data privacy, there is a compelling case for adopting an FL framework. To further address the comprehensive challenges in the realistic IIoT environment characterized by heterogeneous data distribution and limited device availability, a FedIn-NID framework is proposed in Section III.

III. PROPOSED FEDIN-NID FRAMEWORK

In this section, we introduce the proposed FedIn-NID framework. We first provide an overview of the framework. Then we present the multidimensional client selection strategy, followed by the proposed model structure design for network intrusion detection, the descriptions of local training, and the proposed FedEMA aggregation algorithm.

A. FedIn-NID Framework

Fig. 1 and Algorithm 1 provide the overview and procedure of our designed FedIn-NID framework in investigated large-scale heterogeneous IIoT environment. In the initialization phase, the global model is used to initialize the local models of all clients (step ①). In each communication round, client selection is performed to obtain a subset of clients for participating in the current round (step ②). All participating clients undergo local training (step ③), then upload the updated local models to the server (step ④). The server employs our proposed FedEMA to aggregate uploaded local models (step ⑤), resulting in a new global model for the next communication round (step ⑥).

Algorithm 1 FedIn-NID Framework

Input: number of communication rounds $\mathcal{T}$; total clients $\mathcal{C}$; participating client ratio F ; number of local epochs E ; learning rate η ; hyper-parameter λ, β

Output: global model $\theta_g^{\mathcal{T}}$

```

1 Server executes:
2 Initialize  $\theta_g^0$ 
3 for  $t = 0, 1, \dots, \mathcal{T} - 1$  do
4    $m \leftarrow \max(1, \lfloor |\mathcal{C}| * F \rfloor)$ 
5    $\mathcal{C}_{\text{select}}^t, \mathcal{I}_{\text{select}}^t \leftarrow \text{ClientSelect}(\mathcal{C}, m)$  via Algorithm 2
6   for  $\mathcal{C}_i \in \mathcal{C}_{\text{select}}^t$  do in parallel
7     Send the global model  $\theta_g^t$  to  $\mathcal{C}_i$ 
8      $\theta_i^t \leftarrow \text{LocalTraining}(\mathcal{C}_i, \theta_g^t)$ 
9    $\alpha^t \leftarrow \lambda e^{-\beta(1-\frac{t+1}{\mathcal{T}})^2}$ 
10   $\theta_g^{t+1} \leftarrow \alpha^t \theta_g^t + (1 - \alpha^t) \sum_{i \in \mathcal{I}_{\text{select}}^t} \frac{N_i}{\sum_{j \in \mathcal{I}_{\text{select}}^t} N_j} \theta_i^t$ 
11 return  $\theta_g^{\mathcal{T}}$ 
12 LocalTraining $(\mathcal{C}_i, \theta_g^t)$ :
13  $\theta_i^t \leftarrow \theta_g^t$ 
14 for epoch  $e = 1, 2, \dots, E$  do
15   for each batch  $\{\mathcal{X}, \mathcal{Y}\}$  of  $\mathcal{D}_i$  do
16      $\mathcal{L} \leftarrow \mathcal{L}_{CE}(\mathcal{X}, \mathcal{Y})$  via (18)
17      $\theta_i^t \leftarrow \theta_i^t - \eta \nabla \mathcal{L}$ 
18 return  $\theta_i^t$  to server

```

In addition to the aforementioned algorithmic procedure, the framework includes the following pivotal parts: the proposed client selection strategy, the proposed model design structure, the local training method, and the proposed FedEMA global aggregation algorithm. Each of these details will be introduced in the following subsections.

B. Multidimensional Client Selection Strategy

In practice, the limited availability of the devices in the IIoT environment restricts their consistent participation in each round of FL, different from the devices in data centers. The main discrepancy between end devices and data centers is their availability [15], which is affected by the interruption due to changes in device state and network connectivity. Hence, in the communication round of FL (step ③), only a small subset of all devices can participate.

In [17], an initial client selection strategy based on random sampling is proposed. In this approach, a subset of clients is randomly chosen to participate in each communication round of federated learning. Due to its straightforward implementation, this random sampling method has been widely adopted in subsequent works [32]. However, random selection is trivial as it does not account for specific client availabilities. Additionally, it fails to consider the variability and heterogeneity of local datasets among clients. Addressing these factors is essential for the effectiveness and efficiency of federated

Algorithm 2 Multidimensional Client Selection at the t-Th Communication Round

Input: total clients $\mathcal{C}$; number of participating clients m

Output: participating clients $\mathcal{C}_{\text{select}}$, indices of participating clients $\mathcal{I}_{\text{select}}$

```

1 Server executes ClientSelect $(\mathcal{C}, m)$ :
2 for  $i = 1, 2, \dots, |\mathcal{C}|$  do in parallel
3    $\mathcal{N}_i \leftarrow \frac{|\mathcal{C}| * N_i}{\sum_{i=1}^{|\mathcal{C}|} N_i}, \mathcal{B}_i \leftarrow \text{GetScore}(\mathcal{D}_i)$ 
4    $\mathcal{R}_i \leftarrow (\mathcal{A}_i^t)^{w_A} * (\mathcal{N}_i)^{w_N} * (\mathcal{B}_i)^{w_B}$ 
5  $\mathcal{R} = \{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_{|\mathcal{C}|}\}$ 
6  $\mathcal{I} \leftarrow \text{Index}(\mathcal{R}), \mathcal{I} = \{1, 2, \dots, |\mathcal{C}|\}$ 
7  $\mathcal{I}' \leftarrow \text{arg sort}(\mathcal{R}), \mathcal{R}_{\mathcal{I}'_0} \geq \mathcal{R}_{\mathcal{I}'_1} \geq \dots \geq \mathcal{R}_{\mathcal{I}'_{|\mathcal{C}|-1}}$ 
8  $\mathcal{I}_{\text{select}} \leftarrow \{\mathcal{I}'_0, \mathcal{I}'_1, \dots, \mathcal{I}'_{m-1}\}$ 
9  $\mathcal{C}_{\text{select}} \leftarrow \{\mathcal{C}_{\mathcal{I}'_0}, \mathcal{C}_{\mathcal{I}'_1}, \dots, \mathcal{C}_{\mathcal{I}'_{m-1}}\}$ 
10 return  $\mathcal{C}_{\text{select}}, \mathcal{I}_{\text{select}}$ 
11 Client  $\mathcal{C}_i$  executes GetScore $(\mathcal{D}_i)$ :
12 Initialize  $\mathbf{Y}_i \leftarrow \mathbf{0}_K, \mathbf{Y}_i = [Y_{i,0}, Y_{i,1}, \dots, Y_{i,K-1}]$ 
13 for each  $(\mathbf{x}_{i,j}, y_{i,j}) \in \mathcal{D}_i$  do
14    $Y_{i,y_{i,j}} \leftarrow Y_{i,y_{i,j}} + 1$ 
15  $\mathcal{P}_i \leftarrow \frac{\mathbf{Y}_i}{\sum_{k=0}^{K-1} Y_{i,k}}, \mathcal{P}_i = [\mathcal{P}_{i,0}, \mathcal{P}_{i,1}, \dots, \mathcal{P}_{i,K-1}]$ 
16  $\mathcal{H}_i \leftarrow -\sum_{k=0}^{K-1} \mathcal{P}_{i,k} \log_2(\mathcal{P}_{i,k} + \epsilon)$ 
17  $\mathcal{G}_i \leftarrow 1 - \sum_{k=0}^{K-1} \mathcal{P}_{i,k}^2$ 
18  $\mathcal{B}_i \leftarrow \mathcal{H}_i * \mathcal{G}_i$ 
19 return Score  $\mathcal{B}_i$  to server

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learning in large-scale heterogeneous IIoT environments. This becomes even more important when facing diverse network attacks with varying intensities and extreme heterogeneity. To this end, we introduce a novel client selection strategy in our proposed FedIn-NID framework, as illustrated in Algorithm 2.

Firstly, for each client $\mathcal{C}_i$ and its local dataset $\mathcal{D}_i$, the client derives the count $\mathbf{Y}_i$ and distribution probability $\mathcal{P}_i$ of attack categories in $\mathcal{D}_i$ as follows:

$$Y_{i,y_{i,j}} = Y_{i,y_{i,j}} + 1, \quad \text{for each } (\mathbf{x}_{i,j}, y_{i,j}) \in \mathcal{D}_i, \quad (8)$$

$$\mathcal{P}_i = \frac{\mathbf{Y}_i}{\sum_{k=0}^{K-1} Y_{i,k}}, \quad (9)$$

where $Y_{i,k} \in \mathbf{Y}_i = [Y_{i,0}, Y_{i,1}, \dots, Y_{i,K-1}] \in \mathbb{N}^K$ is initialized to 0 before counting attack categories, and $\mathcal{P}_i = [\mathcal{P}_{i,0}, \mathcal{P}_{i,1}, \dots, \mathcal{P}_{i,K-1}] \in \mathbb{R}^K$. Hence, the local dataset distribution of client $\mathcal{C}_i$ can be quantified as follows:

$$\mathcal{H}_i = -\sum_{k=0}^{K-1} \mathcal{P}_{i,k} \log_2(\mathcal{P}_{i,k} + \epsilon), \quad (10)$$

$$\mathcal{G}_i = 1 - \sum_{k=0}^{K-1} \mathcal{P}_{i,k}^2, \quad (11)$$

$$\mathcal{B}_i = \mathcal{H}_i * \mathcal{G}_i = \left(\sum_{k=0}^{K-1} \mathcal{P}_{i,k} \log_2(\mathcal{P}_{i,k} + \epsilon) \right)$$

$$* \left(\sum_{k=0}^{K-1} \mathcal{P}_{i,k}^2 - 1 \right), \quad (12)$$

where $\epsilon = 1e^{-8}$ is a small positive constant, $\mathcal{P}_{i,k} \in \mathcal{P}_i$ is the probability distribution of class k for client $\mathcal{C}_i$, $\mathcal{H}_i$ is the entropy of $\mathcal{P}_i$, $\mathcal{G}_i$ is the Gini coefficient (also known as the Gini index) of $\mathcal{P}_i$, a common metric used to measure the inequality of a distribution [33]. A higher Gini coefficient indicates a greater concentration of data within a few categories, signifying a less balanced distribution. $\mathcal{B}_i$ is the distribution balance factor. The combination of entropy and the Gini coefficient reflects both the balance and uniformity of the data distribution. By multiplying entropy and Gini coefficient, the balance of the client's dataset distribution is quantified. A higher $\mathcal{B}_i$ indicates that the data distribution $\mathcal{P}_i$ of client $\mathcal{C}_i$ is more uniform and balanced, without being dominated by any particular category.

In addition to data distribution, the quantity of local dataset $\mathcal{D}_i$ in each client $\mathcal{C}_i$ also plays a crucial role in FL. Therefore, we define the local dataset volume impact factor for each client as follows:

$$\mathcal{N}_i = \frac{|\mathcal{C}| * N_i}{\sum_{i=1}^{|\mathcal{C}|} N_i}, \quad (13)$$

where $N_i = |\mathcal{D}_i|$ is the quantity of local dataset $\mathcal{D}_i$, and $|\mathcal{C}|$ represents the total number of clients, M . The $|\mathcal{C}|$ term serves to scale this factor by normalizing the client's data contribution against the average data volume across all clients. This ensures $\mathcal{N}_i$ is on a comparable numerical scale to the other factors in (14). Hence, by synergistically considering these multidimensional factors, which do not compromise the data privacy of clients, we ultimately derive the selection weight $\mathcal{R}_i$ for each client $\mathcal{C}_i$ by combining these factors. To allow for tunable importance of each dimension while preserving the multiplicative gating effect, we use an exponential weighting scheme:

$$\mathcal{R}_i = (\mathcal{A}_i')^{w_A} * (\mathcal{N}_i)^{w_N} * (\mathcal{B}_i)^{w_B}, \quad (14)$$

where $\mathcal{R}_i \in \mathcal{R} = \{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_{|\mathcal{C}|}\}$, and the non-negative exponents (w_A, w_N, w_B) control the sensitivity of the total score to each factor. A weight $w_k > 1$ makes the selection criterion stricter by amplifying the influence of the corresponding factor k . This intensifies penalties for clients with suboptimal scores (e.g., in availability) and accentuates differences in other dimensions by strongly rewarding high scores and penalizing low ones (e.g., in data volume and distribution). Conversely, a weight $0 < w_k < 1$ reduces the influence factor k , leading to a more lenient selection by narrowing disparities among clients. This formulation offers flexibility in tuning the selection behavior to suit different operational scenarios. By comprehensively considering client availability, local dataset distribution, and the size of the local dataset, the selection weights $\mathcal{R}_i$ can be used to optimize the selection of participating clients in large-scale heterogeneous IIoT environments. The details are as follows:

$$\mathcal{I} = \text{Index}(\mathcal{R}) = \{1, 2, \dots, |\mathcal{C}|\}, \quad (15)$$

$$\mathcal{I}' = \arg \text{sort}(\mathcal{R}), \text{ where } \mathcal{R}_{\mathcal{I}'_0} \geq \mathcal{R}_{\mathcal{I}'_1} \geq \dots \geq \mathcal{R}_{\mathcal{I}'_{|\mathcal{C}|-1}}, \quad (16)$$

where $\text{Index}(\cdot)$ is the indices function that maps an index to each element in the set, and $\arg \text{sort}(\cdot)$ is the index-sorting

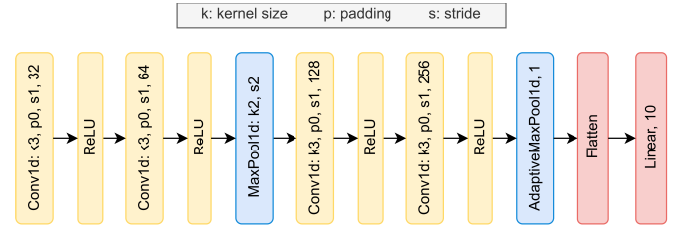


Fig. 2. Structure of designed 1D-CNN for network intrusion detection. The final linear layer outputs predictions for 10 classes, corresponding to one benign class and the nine attack types detailed in Section IV-A.

function that returns the indices of the set sorted in descending order. Therefore, the selected participating clients and their corresponding indices are as follows:

$$\begin{aligned} \mathcal{I}_{\text{select}} &= \{\mathcal{I}'_0, \mathcal{I}'_1, \dots, \mathcal{I}'_{m-1}\}, \\ \mathcal{C}_{\text{select}} &= \{\mathcal{C}_{\mathcal{I}'_0}, \mathcal{C}_{\mathcal{I}'_1}, \dots, \mathcal{C}_{\mathcal{I}'_{m-1}}\}, \end{aligned} \quad (17)$$

where $m = \max(1, \lfloor |\mathcal{C}| * F \rfloor)$ is the number of participating clients, and $F \in [0, 1]$ represents the client participation ratio.

C. Model Structure Design

In the continuation of our FedIn-NID framework, the model structure is intricately designed to cater to the intricate requirements of network intrusion detection within the IIoT landscape. At its core, the model harnesses a deep neural network architecture optimized for extracting meaningful representations from the network traffic data. To effectively classify the attack types of NetFlow traffic for the NID task, we introduce a dedicated 1-dimensional CNN (1D-CNN) model structure tailored specifically for NID.

As depicted in Fig. 2, we employ a 1D-CNN as the feature extractor for the NID task, with a fully connected layer serving as the classification head to output the classification predictions. The designed network architecture primarily consists of 1D convolutional layers, ReLU activation layers, max-pooling layers, and fully connected layers. More specifically, we employ four 1D convolutional layers with a kernel size of 3, no padding, and a stride of 1, with output dimensions of 32, 64, 128, and 256, respectively. Each convolutional layer is subsequently followed by a ReLU activation layer, collectively constituting a convolutional module. Additionally, following the initial two convolutional modules and the latter two, we apply max-pooling and adaptive max-pooling to downsample the convolutional results, respectively. Finally, a fully connected layer is used as the classification head, yielding classification logits as the output.

D. Local Training for NID

For each participating client $\mathcal{C}_i$ in the t -th communication round, we conduct local training for NID on θ_i^t by minimizing the loss as follows:

$$\begin{aligned} \mathcal{L}_{CE}(\mathbf{x}_{i,j}, \mathbf{y}_{i,j}) &= -\frac{1}{N_i} \sum_{j=1}^{N_i} \sum_{k=0}^{K-1} \mathbb{I}(\mathbf{y}_{i,j} = k) \log(\hat{\mathbf{y}}_{i,j})_k, \\ \hat{\mathbf{y}}_{i,j} &= \text{softmax}(f(\mathbf{x}_{i,j}; \theta_i^t)) \end{aligned} \quad (18)$$

$$= \frac{e^{f(\mathbf{x}_{i,j}; \boldsymbol{\theta}_i^t)}}{\sum_k e^{(f(\mathbf{x}_{i,j}; \boldsymbol{\theta}_i^t))_k}}, \quad (19)$$

where $f(\mathbf{x}_{i,j}; \boldsymbol{\theta}_i^t)$ is the output of given network flow $\mathbf{x}_{i,j}$ through a NID model that is parameterized by $\boldsymbol{\theta}_i^t$, $\mathbb{I}(\cdot)$ is the indicator function, and $(\cdot)_k$ denotes the k^{th} element of $(\cdot)$.

E. Global Aggregation With FedEMA

During the global model aggregation at the server side, the most widely employed aggregation method is FedAvg [17]:

$$\boldsymbol{\theta}_g^{t+1} = \sum_{i=1}^n \frac{N_i}{\sum_{j=1}^n N_j} \boldsymbol{\theta}_i^t, \quad (20)$$

where $\boldsymbol{\theta}_i^t$ is the local model of $\mathcal{C}_i$ at the t -th communication round and $\boldsymbol{\theta}_g^{t+1}$ is the global model for the next round.

However, naively aggregating local models using FedAvg makes it challenging to effectively adapt to the heterogeneous and dynamic NID environment in federated learning. Furthermore, in the next-generation IIoT environments with large-scale clients, traditional solutions for heterogeneous federated learning struggle to address the challenges posed by massive clients combined with a low participation ratio. It is difficult to strike an effective balance between the overall global model and the local models acquired by a small subset of participating clients. Consequently, straightforward application of FedAvg can easily lead to the global model bias, hindering the acquisition of genuinely valuable knowledge.

To address this, we propose a novel dynamic aggregation strategy, termed FedEMA. By incorporating the concept of Exponential Moving Average, FedEMA dynamically balances the relatively older but holistic knowledge acquired by the global model with the relatively newer but partial knowledge acquired by a small subset of participating clients. This approach enables us to encompass both previously learned global knowledge and newly acquired local knowledge. Specifically, we calculate the global model for the next communication round using FedEMA as follows:

$$\boldsymbol{\theta}_g^{t+1} = \alpha \boldsymbol{\theta}_g^t + (1 - \alpha) \sum_{i=1}^n \frac{N_i}{\sum_{j=1}^n N_j} \boldsymbol{\theta}_i^t, \quad (21)$$

$$\alpha = \lambda e^{-\beta(1 - \frac{t+1}{T})^2}, \quad (22)$$

where α is the exponential moving average factor, FedEMA degenerates into FedAvg when α is 0 and retains the global model from the previous round without any updates when α is 1; $\lambda \in [0, 1]$ is the extremum control coefficient; $\beta > 0$ is the control coefficient of the exponential moving average process; T is the total number of communication round.

IV. EXPERIMENT AND RESULTS

Given the requirements for data security and privacy in the large-scale heterogeneous IIoT environment, effectively conducting network intrusion detection under the premise of privacy protection is crucial. Furthermore, the fluctuations in device statuses, operating system quotas, and network connections across a myriad of IIoT devices result in device interruptions, leading to limited device availability. This makes

it exceedingly challenging for all devices to maintain a consistent active state during each communication round. Consequently, the central server can only gather responses from a small fraction of devices for updates, causing the global model bias and deviating toward different subsets of devices during the iterative training process, diminishing convergence speed and model performance. To this end, we set a vast number of clients within the federated learning scenario (ranging from 50 to 500), but with a low client participation ratio (10%), to simulate the realistic IIoT environment. To demonstrate the performance and robustness of our FedIn-NID in such a realistic IIoT scenario, we conduct experiments on a real IoT dataset and further evaluate our approach with elaborate settings.

A. Dataset and Preprocessing

We employ NF-ToN-IoT-v2 [34] as the network intrusion detection dataset, which is an IoT network dataset based on NetFlow involving diverse attack categories, and its NetFlow records are generated using publicly available pcap files from the ToN-IoT dataset. The dataset comprises a total of 16,940,496 data flows, with 10,841,027 (63.99%) attack instances and 6,099,469 (36.01%) benign instances. Specifically, the attack categories consist of nine distinct types: Scanning, XSS, DDoS, Password, DoS, Injection, Backdoor, MITM, and Ransomware. Besides, each data flow is made up of 43 NetFlow features.

We perform preprocessing for this dataset at the experimental setup stage. To begin with, we remove two features in the dataset: `IPV4_SRC_ADDR` and `IPV4_DST_ADDR`. Subsequently, we perform label encoding to all remaining features and drop the redundant data flows. Finally, we randomly split 60% of data flows as `train` and the rest of 40% as `test`, followed by dataset normalization.

We conduct data partitions on `train` according to Section II-A, Figs. 3, 4 and 5 illustrate the data distributions under varying Non-IID settings with $\text{Dir}(\gamma)$ for diverse numbers of clients. It is evident that as γ decreases from 0.5 to 0.2 and then to 0.1, the heterogeneity in data distribution significantly intensifies. This strategy of partitioning a feature-rich dataset across numerous, heterogeneously distributed, and intermittently available clients (as detailed in Section IV-C) allows us to effectively simulate the complex data and device landscape of a real-world large-scale heterogeneous IIoT environment.

B. Compared Methods

We compare our method against three methods, including:

- 1) FedAvg [17], which serves as a baseline for federated learning;
- 2) FedProx [35], which incorporates a proximal term during local model training to prevent over-bias, becoming the baseline for federated learning under Non-IID data settings;
- 3) SCAFFOLD [32], which corrects the overall model training direction using a control variable, standing as a state-of-the-art for federated learning under Non-IID data settings.

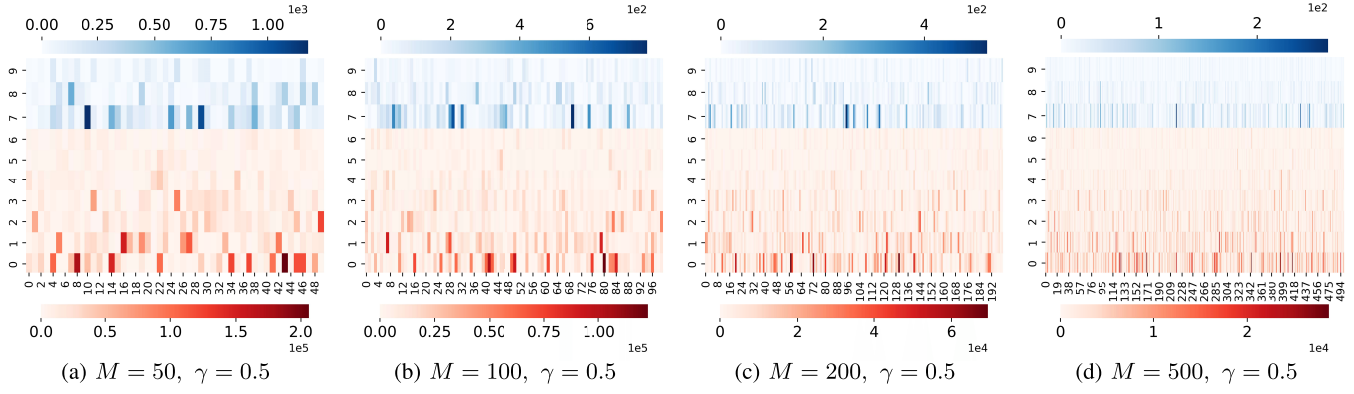


Fig. 3. Data distribution of each client under Non-IID setting with Dir(0.5). The color bar refers to the number of data samples. Each patch represents the number of data samples for a specific class in a client. The horizontal axis denotes the client ID, and the vertical axis denotes the attack class ID.

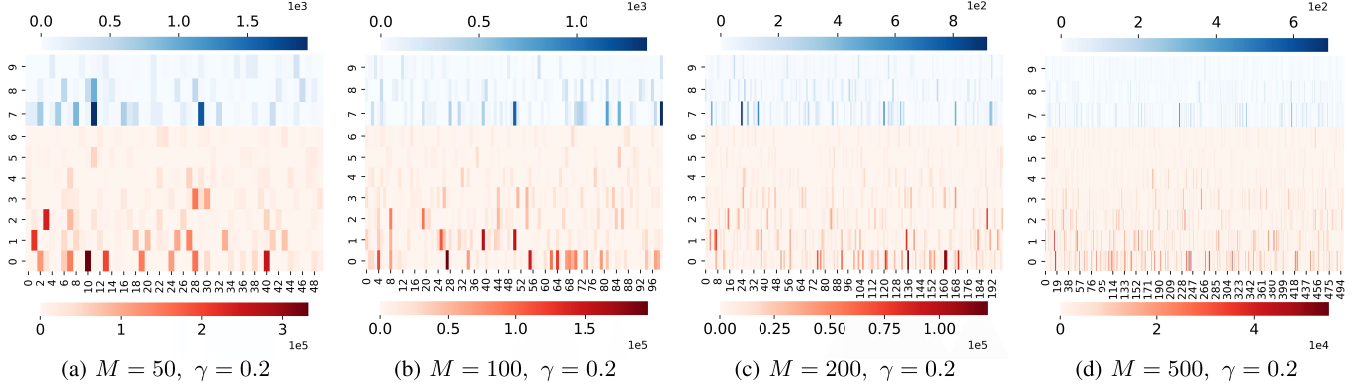


Fig. 4. Data distribution of each client under Non-IID setting with Dir(0.2). The color bar refers to the number of data samples. Each patch represents the number of data samples for a specific class in a client. The horizontal axis denotes the client ID, and the vertical axis denotes the attack class ID.

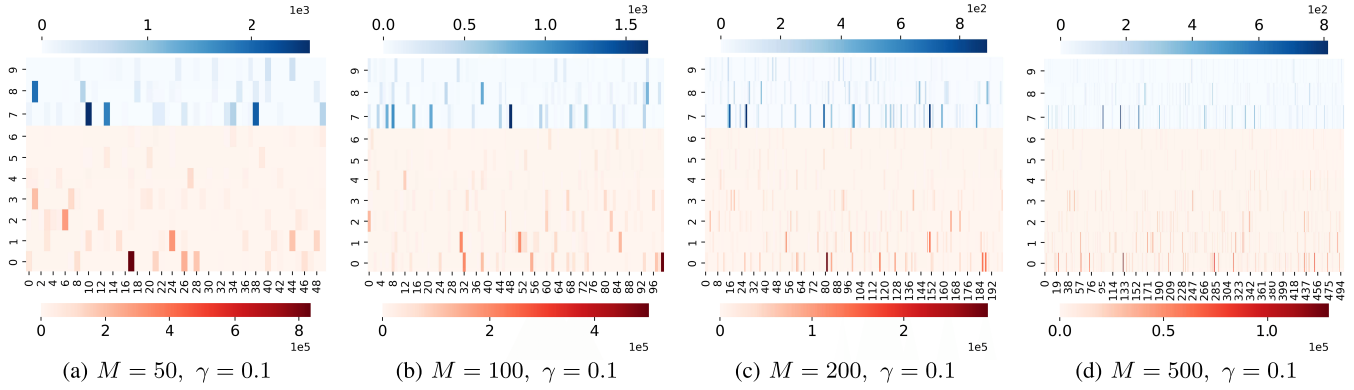


Fig. 5. Data distribution of each client under Non-IID setting with Dir(0.1). The color bar refers to the number of data samples. Each patch represents the number of data samples for a specific class in a client. The horizontal axis denotes the client ID, and the vertical axis denotes the attack class ID.

To more accurately evaluate the performance of each baseline in large-scale heterogeneous IIoT environments, we employ a client selection strategy based on client availability for these methods as follows:

$$\mathcal{I}^\circ = \arg \text{sort}(\mathcal{A}^t), \text{ where } \mathcal{A}_{T_0}^t \geq \mathcal{A}_{T_1}^t \geq \dots \geq \mathcal{A}_{T_{|\mathcal{I}^\circ|-1}}^t, \quad (23)$$

$$\mathcal{I}_{\text{select}}^\circ = \{\mathcal{I}_0^\circ, \mathcal{I}_1^\circ, \dots, \mathcal{I}_{m-1}^\circ\},$$

$$\mathcal{C}_{\text{select}}^\circ = \{\mathcal{C}_{\mathcal{I}_0^\circ}, \mathcal{C}_{\mathcal{I}_1^\circ}, \dots, \mathcal{C}_{\mathcal{I}_{m-1}^\circ}\}, \quad (24)$$

where $\mathcal{A}_i^t \in \mathcal{A}^t = \{\mathcal{A}_1^t, \mathcal{A}_2^t, \dots, \mathcal{A}_{|\mathcal{C}|}^t\}$ is the device availability for client $\mathcal{C}_i$ at the t -th communication round, and $\mathcal{C}_{\text{select}}^\circ$ is the selected clients for these baseline methods at this round.

C. Implementation Details

We employ Pytorch to conduct our experiment, utilizing the Adam optimizer with an initial learning rate of $1e^{-3}$. The batch size is set to 1024. We train 50 communication rounds ($\mathcal{T}$), with the client participation ratio (F) of 0.1, and the local epoch (E) is set to 1. During the training stage, we adopt AMP (Automatic mixed precision) to all methods. Evaluations are conducted at an interval of 5 communication rounds. For two hyper-parameters of FedEMA, we empirically set the extremum control coefficient $\lambda = 0.5$ and the control coefficient of the exponential moving average process $\beta = 4.0$ for all experiments. Furthermore, we set $w_{\mathcal{A}} = w_{\mathcal{N}} = w_{\mathcal{B}} = 1$

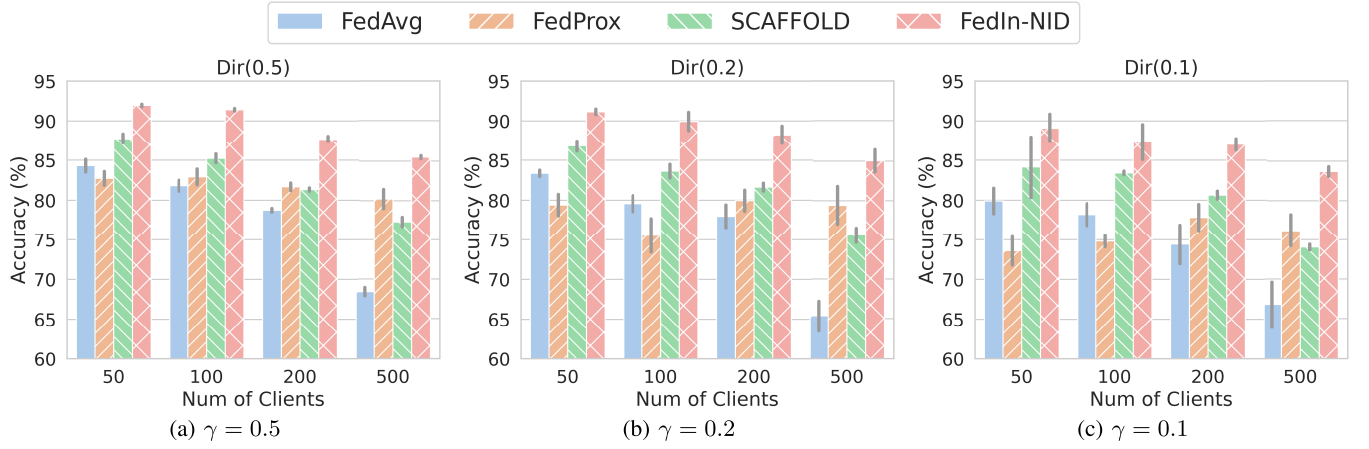


Fig. 6. Illustrate the quantitative results (average and standard deviation of 5 trials) on test in terms of accuracy.

TABLE I
QUANTITATIVE RESULTS (AVERAGE AND STANDARD DEVIATION OF 5 TRIALS) ON TEST IN TERMS OF F1-SCORE UNDER DIR(0.5) SETTING

Method	Benign	Scanning	XSS	DDoS	Password	DoS	Injection	Backdoor	MITM	Ransomware	macro-average	IMP
M = 50												
FedAvg	94.07±0.56	95.04±0.33	72.71±1.01	87.56±2.15	65.47±0.66	24.67±1.24	45.77±6.59	97.48±0.90	0.00±0.00	0.00±0.00	58.28±1.02	—
FedProx	93.91±0.68	94.84±0.52	67.14±2.99	82.47±3.18	63.77±5.60	46.29±16.70	33.82±9.61	93.40±3.72	0.00±0.00	0.00±0.00	57.56±0.82	↓ 0.71±0.91
SCAFFOLD	95.62±0.04	95.99±0.01	79.49±0.99	93.00±0.06	70.64±3.70	42.62±10.97	49.91±0.41	98.63±0.12	1.60±2.22	48.86±30.81	67.63±3.77	↑ 9.36±2.76
FedIn-NID (ours)	96.91±0.69	96.35±0.61	88.99±2.28	94.51±0.12	83.27±0.39	74.84±7.90	52.79±7.65	99.06±0.10	9.76±1.59	75.16±3.64	77.16±0.39	↑ 18.88±1.22
M = 100												
FedAvg	93.72±0.40	95.01±0.26	66.88±0.46	80.18±1.56	63.01±1.91	23.72±0.23	12.78±10.81	98.52±0.08	0.00±0.00	0.00±0.00	53.38±1.24	—
FedProx	94.16±0.65	94.76±0.86	66.31±2.17	84.82±2.76	60.47±2.52	47.14±16.22	41.61±6.53	96.18±2.57	0.00±0.00	0.00±0.00	58.54±1.97	↑ 5.16±3.19
SCAFFOLD	94.98±0.22	95.72±0.16	72.59±0.75	90.20±0.67	64.46±2.09	26.36±1.29	36.59±8.26	97.69±0.48	0.00±0.00	0.94±1.33	57.95±1.10	↑ 4.57±0.50
FedIn-NID (ours)	97.21±0.04	96.95±0.07	86.71±0.48	94.18±0.07	82.62±0.72	63.40±0.50	52.07±0.35	98.96±0.22	0.78±1.11	67.76±0.69	74.06±0.28	↑ 20.68±1.01
M = 200												
FedAvg	91.78±0.11	93.86±0.08	62.89±0.35	74.63±0.93	60.77±0.52	13.77±1.06	0.00±0.00	24.83±35.11	0.00±0.00	0.00±0.00	42.25±3.40	—
FedProx	93.74±0.37	94.70±0.47	61.90±5.09	81.57±2.08	62.55±2.21	47.42±12.90	25.01±7.15	97.47±0.50	0.00±0.00	0.00±0.00	56.44±1.30	↑ 14.18±4.49
SCAFFOLD	94.98±0.08	95.12±0.06	65.64±0.49	79.42±0.39	63.10±0.34	23.31±0.94	0.75±0.79	96.40±2.44	0.00±0.00	0.00±0.00	51.76±0.19	↑ 9.51±3.43
FedIn-NID (ours)	95.93±0.08	96.41±0.13	78.63±0.53	92.50±0.65	71.33±2.22	24.06±0.90	43.75±10.56	98.51±0.02	0.00±0.00	23.73±30.72	62.49±3.54	↑ 20.23±6.50
M = 500												
FedAvg	77.44±0.98	75.33±0.59	59.11±0.35	71.85±0.39	55.33±0.10	12.69±0.33	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	35.17±0.19	—
FedProx	93.52±0.55	94.77±0.46	64.05±1.08	74.48±2.16	62.04±3.26	32.15±13.39	28.90±9.33	0.00±0.00	0.00±0.00	0.00±0.00	44.99±1.11	↑ 9.82±1.27
SCAFFOLD	90.25±0.08	91.67±0.77	62.37±0.28	73.48±0.37	57.99±1.10	20.42±4.07	0.05±0.07	0.00±0.00	0.00±0.00	0.00±0.00	39.62±0.59	↑ 4.45±0.55
FedIn-NID (ours)	94.95±0.04	95.83±0.04	72.67±1.02	89.80±0.80	67.70±3.11	35.05±17.50	21.76±1.79	98.48±0.15	0.00±0.00	0.00±0.00	57.62±1.82	↑ 22.45±1.92

to FedIn-NID. To reduce computational complexity, the availability of each client is sampled from a random distribution in each communication round.

D. Evaluation Metrics

To evaluate the performance of network intrusion detection, we employ widely-used metrics such as F1-score and accuracy for the multi-class classification task. Furthermore, we utilize the macro-average to calculate average values for the multi-class NID task.

1) *F1-Score*: This metric represents the harmonic average of precision and recall, formulated as follows:

$$\begin{aligned} F1 &= \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \\ &= \frac{2 * TP}{2 * TP + FN + FP}. \end{aligned} \quad (25)$$

This metric provides a balanced measure of accuracy for the NID task. We calculate the macro-F1 as follows:

$$\text{macro-F1} = \frac{1}{K} \sum_{k=0}^{K-1} F1^{(k)}, \quad (26)$$

where K is the number of classes, and $F1^{(k)}$ denotes the F1-score for the k^{th} class.

2) *Accuracy*: The multi-class accuracy is given by:

$$\text{Accuracy} = \frac{\sum_{k=0}^{K-1} TP^{(k)}}{\sum_{k=0}^{K-1} (TP^{(k)} + FN^{(k)})}, \quad (27)$$

where K is the number of classes, $TP^{(k)}$ and $FN^{(k)}$ denote the true positive and false negative values for the k^{th} class in the confusion matrix, respectively.

E. Quantitative Results

Tables I, II, and III present the quantitative results of our proposed FedIn-NID and other methods on NF-ToN-IoT-v2 in terms of F1. The IMP column in the tables demonstrates the improvement of the other methods over the traditional FedAvg. Fig. 6, illustrates the quantitative results of our proposed FedIn-NID and other methods on NF-ToN-IoT-v2 in terms of Accuracy. Due to the diversification of NetFlow attack types and the disparities in encountering different network attacks, the NetFlow attack category classification datasets often exhibit severe class imbalance. For instance, the benign category constitutes the majority, while there are significant numerical disparities among attack categories. Therefore, we comprehensively evaluate performance employing the F1 and

TABLE II
QUANTITATIVE RESULTS (AVERAGE AND STANDARD DEVIATION OF 5 TRIALS) ON **TEST** IN TERMS OF F1-SCORE UNDER DIR(0.2) SETTING

Method	Benign	Scanning	XSS	DDoS	Password	DoS	Injection	Backdoor	MITM	Ransomware	macro-average	IMP
$M = 50$												
FedAvg	94.21 $\pm$ 0.11	94.81 $\pm$ 0.31	70.14 $\pm$ 1.22	84.14 $\pm$ 1.42	61.42 $\pm$ 5.10	42.63 $\pm$ 14.67	27.70 $\pm$ 19.78	91.23 $\pm$ 2.88	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	56.63 $\pm$ 3.27	—
FedProx	93.10 $\pm$ 0.85	93.60 $\pm$ 0.39	55.94 $\pm$ 6.95	79.43 $\pm$ 4.26	58.84 $\pm$ 2.66	46.06 $\pm$ 6.98	20.63 $\pm$ 14.68	91.23 $\pm$ 5.72	0.00 $\pm$ 0.00	1.04 $\pm$ 1.47	53.99 $\pm$ 2.94	$\downarrow$ 2.64 $\pm$ 4.36
SCAFFOLD	95.36 $\pm$ 0.45	95.48 $\pm$ 0.43	75.12 $\pm$ 2.78	92.02 $\pm$ 0.95	73.45 $\pm$ 1.49	50.56 $\pm$ 14.72	39.70 $\pm$ 7.49	95.49 $\pm$ 2.82	3.86 $\pm$ 5.46	41.57 $\pm$ 29.58	66.26 $\pm$ 2.74	$\uparrow$ 9.63 $\pm$ 4.50
FedIn-NID (ours)	97.48$\pm$0.15	97.08$\pm$0.10	87.45$\pm$0.63	92.82$\pm$1.09	80.51$\pm$3.04	62.77$\pm$0.41	32.54$\pm$1.79	99.17$\pm$0.03	0.00$\pm$0.00	0.00$\pm$0.00	64.98$\pm$0.41	$\uparrow$ 8.35$\pm$3.01
$M = 100$												
FedAvg	92.98 $\pm$ 0.97	94.24 $\pm$ 0.62	63.20 $\pm$ 0.25	74.00 $\pm$ 1.16	55.56 $\pm$ 5.62	22.89 $\pm$ 0.92	0.00 $\pm$ 0.00	94.60 $\pm$ 4.30	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	49.75 $\pm$ 0.56	—
FedProx	93.37 $\pm$ 0.59	94.03 $\pm$ 1.01	27.83 $\pm$ 24.78	76.78 $\pm$ 1.14	54.21 $\pm$ 2.21	39.10 $\pm$ 11.61	25.62 $\pm$ 4.60	95.12 $\pm$ 3.05	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	50.61 $\pm$ 1.28	$\uparrow$ 0.86 $\pm$ 1.71
SCAFFOLD	94.30 $\pm$ 0.17	95.30 $\pm$ 0.36	69.48 $\pm$ 2.02	85.17 $\pm$ 3.41	63.74 $\pm$ 1.27	27.93 $\pm$ 2.66	22.58 $\pm$ 2.63	94.37 $\pm$ 2.45	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	55.29 $\pm$ 0.16	$\uparrow$ 5.54 $\pm$ 0.42
FedIn-NID (ours)	96.68$\pm$0.20	95.83$\pm$0.46	85.61$\pm$2.62	93.22$\pm$1.68	61.94$\pm$4.84	80.78$\pm$5.53	53.73$\pm$4.64	98.66$\pm$0.02	0.17$\pm$0.24	70.27$\pm$2.31	73.69$\pm$1.79	$\uparrow$ 23.94$\pm$1.76
$M = 200$												
FedAvg	90.21 $\pm$ 0.97	92.66 $\pm$ 0.66	63.48 $\pm$ 1.27	75.83 $\pm$ 2.62	57.73 $\pm$ 1.40	13.60 $\pm$ 0.77	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	39.35 $\pm$ 0.55	—
FedProx	92.89 $\pm$ 0.83	94.27 $\pm$ 1.01	61.50 $\pm$ 5.98	76.02 $\pm$ 0.89	63.54 $\pm$ 0.34	19.35 $\pm$ 6.45	31.11 $\pm$ 3.29	92.80 $\pm$ 2.65	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	53.15 $\pm$ 1.39	$\uparrow$ 8.44 $\pm$ 1.90
SCAFFOLD	93.84 $\pm$ 0.18	95.01 $\pm$ 0.15	66.22 $\pm$ 1.06	80.92 $\pm$ 2.01	63.18 $\pm$ 0.65	22.65 $\pm$ 0.73	0.59 $\pm$ 0.79	98.37 $\pm$ 0.17	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	52.08 $\pm$ 0.26	$\uparrow$ 12.73 $\pm$ 0.32
FedIn-NID (ours)	96.36$\pm$0.17	96.45$\pm$0.12	79.73$\pm$2.60	91.25$\pm$0.53	75.23$\pm$2.88	39.66$\pm$16.14	25.55$\pm$0.24	98.51$\pm$0.01	0.63$\pm$0.90	45.06$\pm$31.75	64.84$\pm$1.09	$\uparrow$ 25.49$\pm$0.99
$M = 500$												
FedAvg	74.39 $\pm$ 1.29	69.72 $\pm$ 4.05	55.28 $\pm$ 2.66	70.84 $\pm$ 0.18	55.25 $\pm$ 0.06	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	32.55 $\pm$ 0.78	—
FedProx	92.83 $\pm$ 0.52	94.09 $\pm$ 1.15	64.41 $\pm$ 3.12	73.64 $\pm$ 7.37	57.33 $\pm$ 3.29	12.12 $\pm$ 9.46	15.49 $\pm$ 13.71	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	40.99 $\pm$ 1.28	$\uparrow$ 8.44 $\pm$ 2.56
SCAFFOLD	87.88 $\pm$ 0.69	87.76 $\pm$ 1.01	62.44 $\pm$ 0.73	73.65 $\pm$ 1.25	57.26 $\pm$ 0.60	19.54 $\pm$ 3.55	0.12 $\pm$ 0.09	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.86 $\pm$ 0.63	$\uparrow$ 6.32 $\pm$ 1.39
FedIn-NID (ours)	94.86$\pm$0.22	95.97$\pm$0.05	73.05$\pm$3.14	89.76$\pm$2.73	63.18$\pm$5.43	20.83$\pm$1.87	26.64$\pm$9.89	98.46$\pm$0.12	0.00$\pm$0.00	0.00$\pm$0.00	56.28$\pm$1.01	$\uparrow$ 23.73$\pm$1.79

TABLE III
QUANTITATIVE RESULTS (AVERAGE AND STANDARD DEVIATION OF 5 TRIALS) ON **TEST** IN TERMS OF F1-SCORE UNDER DIR(0.1) SETTING

Method	Benign	Scanning	XSS	DDoS	Password	DoS	Injection	Backdoor	MITM	Ransomware	macro-average	IMP
$M = 50$												
FedAvg	89.51 $\pm$ 1.87	90.03 $\pm$ 2.54	71.66 $\pm$ 1.73	82.83 $\pm$ 4.36	60.23 $\pm$ 2.85	15.03 $\pm$ 10.68	46.39 $\pm$ 1.64	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	45.57 $\pm$ 1.67	—
FedProx	87.24 $\pm$ 2.90	88.99 $\pm$ 2.95	59.80 $\pm$ 1.65	62.75 $\pm$ 1.06	55.53 $\pm$ 0.18	21.88 $\pm$ 1.93	8.46 $\pm$ 1.96	86.10 $\pm$ 9.66	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	47.07 $\pm$ 0.10	$\uparrow$ 1.50 $\pm$ 1.72
SCAFFOLD	92.46 $\pm$ 2.80	92.71 $\pm$ 2.77	72.85 $\pm$ 6.39	91.87 $\pm$ 0.89	68.00 $\pm$ 5.91	37.98 $\pm$ 14.31	44.98 $\pm$ 1.40	98.37 $\pm$ 0.21	2.27 $\pm$ 3.21	30.48 $\pm$ 29.45	63.20 $\pm$ 2.86	$\uparrow$ 17.63 $\pm$ 4.53
FedIn-NID (ours)	96.48$\pm$0.45	95.39$\pm$0.79	78.38$\pm$4.03	93.40$\pm$1.01	82.45$\pm$5.55	52.39$\pm$16.19	57.75$\pm$11.96	97.69$\pm$1.00	0.00$\pm$0.00	4.53$\pm$6.41	65.84$\pm$1.13	$\uparrow$ 20.28$\pm$1.21
$M = 100$												
FedAvg	90.08 $\pm$ 1.10	92.44 $\pm$ 1.37	65.30 $\pm$ 1.02	78.31 $\pm$ 3.61	50.11 $\pm$ 2.92	0.00 $\pm$ 0.00	4.38 $\pm$ 6.19	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.06 $\pm$ 0.98	—
FedProx	89.88 $\pm$ 1.70	90.64 $\pm$ 4.32	55.09 $\pm$ 7.91	67.31 $\pm$ 7.20	52.30 $\pm$ 3.29	24.59 $\pm$ 20.66	5.72 $\pm$ 8.09	86.50 $\pm$ 8.50	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	47.20 $\pm$ 1.56	$\uparrow$ 9.14 $\pm$ 0.61
SCAFFOLD	93.01 $\pm$ 0.59	94.80 $\pm$ 0.30	71.21 $\pm$ 0.37	88.96 $\pm$ 1.38	61.45 $\pm$ 3.65	19.31 $\pm$ 4.23	20.18 $\pm$ 14.94	96.39 $\pm$ 1.47	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	54.53 $\pm$ 1.04	$\uparrow$ 16.47 $\pm$ 0.57
FedIn-NID (ours)	96.04$\pm$0.30	95.19$\pm$1.41	82.33$\pm$1.91	93.17$\pm$2.22	49.08$\pm$29.31	44.45$\pm$16.36	41.38$\pm$5.20	97.82$\pm$1.09	8.94$\pm$9.14	76.33$\pm$5.71	68.47$\pm$5.08	$\uparrow$ 30.41$\pm$5.05
$M = 200$												
FedAvg	87.60 $\pm$ 2.50	93.30 $\pm$ 0.93	50.85 $\pm$ 0.07	71.79 $\pm$ 2.67	55.37 $\pm$ 0.79	0.00 $\pm$ 0.00	19.09 $\pm$ 13.50	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	37.80 $\pm$ 0.32	—
FedProx	91.07 $\pm$ 2.45	87.39 $\pm$ 4.11	63.88 $\pm$ 3.11	79.50 $\pm$ 2.84	53.83 $\pm$ 0.35	17.89 $\pm$ 12.86	25.38 $\pm$ 18.46	86.06 $\pm$ 4.97	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	50.50 $\pm$ 1.67	$\uparrow$ 12.70 $\pm$ 1.64
SCAFFOLD	93.03 $\pm$ 0.63	94.12 $\pm$ 0.56	65.57 $\pm$ 0.81	77.18 $\pm$ 1.33	63.02 $\pm$ 0.55	22.39 $\pm$ 0.58	22.96 $\pm$ 17.10	96.47 $\pm$ 1.33	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	53.48 $\pm$ 1.54	$\uparrow$ 15.68 $\pm$ 1.84
FedIn-NID (ours)	95.28$\pm$0.13	95.02$\pm$0.66	78.64$\pm$2.26	92.00$\pm$1.03	71.39$\pm$3.98	21.23$\pm$1.79	52.48$\pm$4.30	98.54$\pm$0.04	0.00$\pm$0.00	24.32$\pm$30.35	62.89$\pm$3.50	$\uparrow$ 25.09$\pm$3.77
$M = 500$												
FedAvg	76.54 $\pm$ 2.61	71.01 $\pm$ 4.86	56.34 $\pm$ 3.32	71.57 $\pm$ 0.53	55.24 $\pm$ 0.06	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	33.07 $\pm$ 1.07	—
FedProx	90.63 $\pm$ 1.04	91.96 $\pm$ 1.97	62.92 $\pm$ 1.53	67.05 $\pm$ 3.18	52.34 $\pm$ 3.93	15.99 $\pm$ 3.16	8.30 $\pm$ 11.73	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	39.48 $\pm$ 1.10	$\uparrow$ 6.41 $\pm$ 2.15
SCAFFOLD	85.92 $\pm$ 0.64	85.03 $\pm$ 0.78	62.04 $\pm$ 0.70	72.99 $\pm$ 1.04	56.59 $\pm$ 0.90	19.99 $\pm$ 3.64	0.02 $\pm$ 0.03	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	38.26 $\pm$ 0.47	$\uparrow$ 5.19 $\pm$ 1.48
FedIn-NID (ours)	94.49$\pm$0.36	94.70$\pm$1.04	70.84$\pm$1.56	86.82$\pm$0.88	62.05$\pm$4.23	31.59$\pm$12.64	17.50$\pm$17.18	98.31$\pm$0.12	0.00$\pm$0.00	0.00$\pm$0.00	55.63$\pm$0.64	$\uparrow$ 22.56$\pm$1.62

Accuracy from fine-grained and coarse-grained perspectives, respectively.

When interpreting the results, it is important to consider the nature of the macro-average F1-score in the context of the highly imbalanced NF-ToN-IIoT-v2 dataset. The macro-average F1-score computes the unweighted mean of per-class F1-scores, giving equal importance to both populous classes (e.g., Benign) and extremely rare attack classes (e.g., Ransomware, MITM). As shown in Tables I, II, and III, achieving a high F1-score on these rare categories is exceptionally challenging, with many baseline methods scoring 0.00. These low scores from the minority classes significantly pull down the overall macro-average value for all methods. The key strength of FedIn-NID is its superior performance on these specific rare classes, which demonstrates its enhanced robustness against severe class imbalance and is the primary driver of its overall performance gains. We analyze the experimental result from two distinct perspectives as follows:

1) *Evaluation on Data Distribution*: It is evident that as the heterogeneity of the datasets increases (i.e., as γ decreases in $\text{Dir}(\gamma)$), the performance of all methods suffers degradation, further demonstrating the impact of heterogeneous data on federated learning [15], [36]. Nevertheless, our FedIn-NID continues to demonstrate outstanding performance and robustness, particularly in small-sample categories such as

Injection, Backdoor, and Ransomware, which become increasingly difficult for other methods to classify as dataset heterogeneity increases. Significantly, FedAvg fails to classify some small-sample categories under all settings, which is primarily due to the fact that our FL scenario involves a vast number of clients, but only a minuscule proportion of clients participate in each communication round. The scenario stands in stark contrast with the generic scenario of FedAvg (with fewer clients and higher participation proportion), further underscoring that FedAvg fails to meet the requirements of the more complex and realistic IIoT environment characterized by data heterogeneity, massive devices, and limited device availability.

From the fine-grained perspective of F1, FedIn-NID outperforms FedAvg by approximately 8% to 30% under all γ settings. For example, under the $\text{Dir}(0.1)$ setting in Table III, FedIn-NID achieves around 20% to 30% improvement over FedAvg in macro-F1, demonstrating its robustness even in extreme heterogeneity scenarios. This further demonstrates that the multidimensional client selection strategy within the FedIn-NID framework ensures efficient and balanced client participation, thus effectively mitigating the heterogeneity in attack data. Other methods, such as FedProx and SCAFFOLD, struggle to maintain high performance as γ decreases, particularly in small-sample categories. The stability and robustness

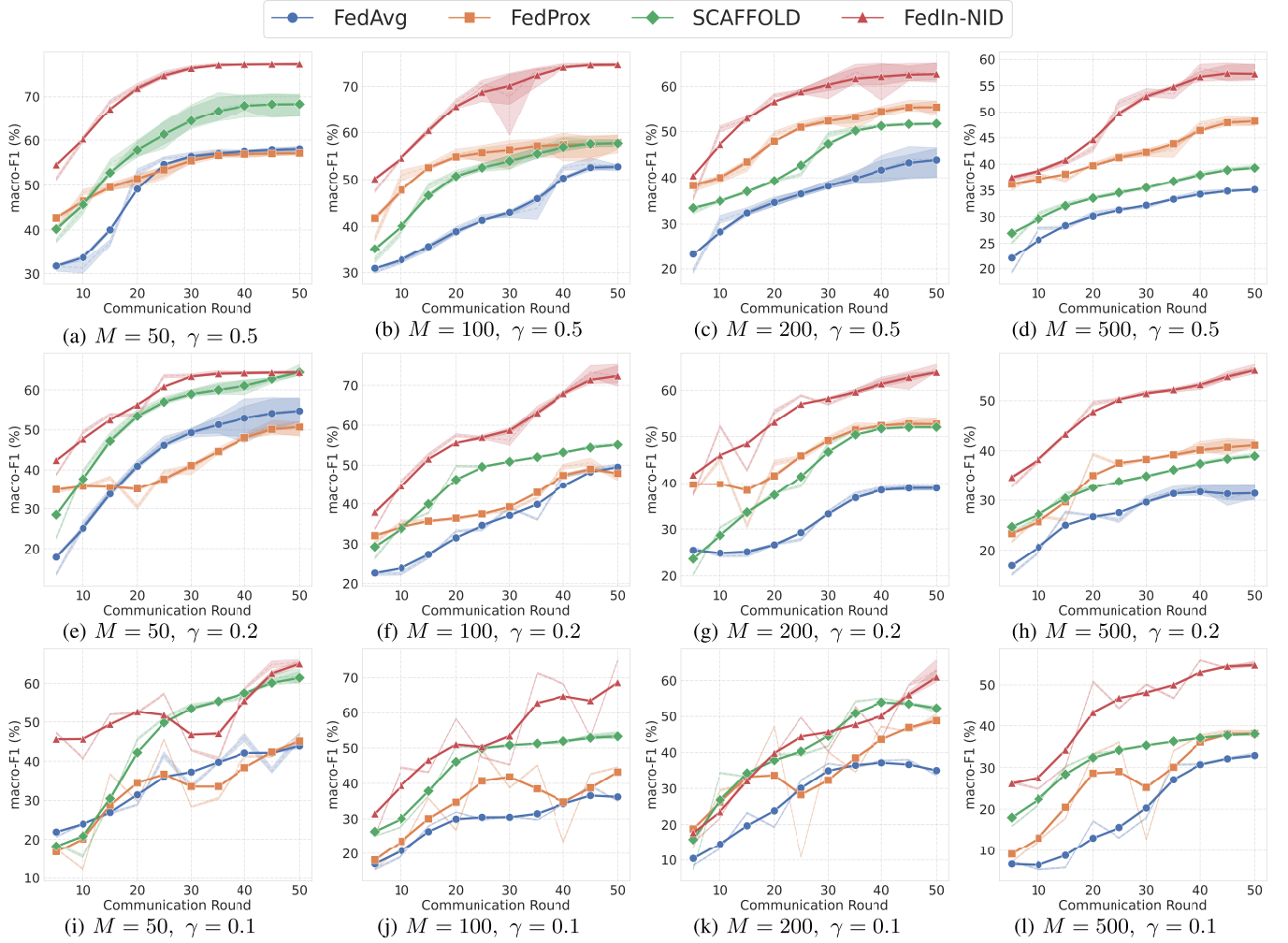


Fig. 7. Convergence plots in terms of macro-F1 for FedIn-NID and baselines. Subplots represent varying total client counts (M) and data heterogeneity levels controlled by the Dirichlet parameter (γ), where a lower γ indicates higher heterogeneity.

of FedIn-NID in classifying both frequent and rare categories make it especially suitable for highly heterogeneous IIoT environments.

From the coarse-grained perspective of Accuracy, FedIn-NID also outperforms other methods, with an average improvement of around 5% to 20% across all γ settings. However, due to the dominance of benign categories, the Accuracy improvements are less pronounced than the F1 improvements. This highlights the need to evaluate performance using multiple metrics, especially in imbalanced classification tasks like network intrusion detection.

2) *Evaluation on the Total Number of Clients:* With an increase in the total number of clients, it is evident that all methods undergo a performance decline. This is attributed to the reduction in the number of training instances per client as the number of total clients grows. When conducting local training with fewer training samples, it becomes easier for the model to overfit, leading to increased disparities among models during global model aggregation, exacerbating model drift. This is due to the reduction in training samples per client, leading to overfitting during local training and increased model drift during global aggregation. However, FedIn-NID consistently demonstrates superior performance across

different client numbers. For example, in the $M = 500$ setting, FedIn-NID delivers a significant improvement over FedAvg, achieving up to 22% higher macro-F1 under the Dir(0.5) and Dir(0.2) settings, and maintaining a 23% improvement even under the more challenging Dir(0.1) setting. This further demonstrates that the FedEMA global model aggregation strategy dynamically balances global model updates by using an exponential moving average to reduce the bias introduced by large-scale and highly diverse client models.

From the coarse-grained perspective of Accuracy, it can be observed that when the total number of clients is relatively low, the performance of FedProx is inferior to FedAvg, particularly under more heterogeneous settings like Dir(0.2) and Dir(0.1). Conversely, with a relatively larger total number of clients, FedProx shows performance improvement against SCAFFOLD. This demonstrates that SCAFFOLD performs worse under the more large-scale client environment. In contrast, FedIn-NID exhibits consistently stable and outstanding performance across all client numbers and data distributions, reinforcing its robustness and suitability for large-scale IIoT.

3) *Convergence Analysis:* To further illustrate the convergence dynamics and the impact of our proposed framework, Fig. 7 presents the macro-F1 score of all methods over the

50 communication rounds for each experimental setting. To better visualize the underlying convergence trends, the solid lines show the average performance smoothed with a Gaussian filter [37] (with $\sigma = 0.9$), while the shaded regions represent the standard deviation of the raw results across trials. The convergence plots clearly show that FedIn-NID consistently achieves a higher macro-F1 score throughout the training process compared to the baseline methods. It not only converges to a superior final performance but also demonstrates a faster convergence rate. This advantage is especially pronounced in the more challenging settings with higher data heterogeneity (i.e., lower γ) and a larger number of clients. This empirically demonstrates that the proposed multidimensional client selection and FedEMA aggregation strategies work in synergy to create a more stable and efficient training process, leading to a more robust final global model.

In summary, the comprehensive experimental results, conducted under varying dataset distributions and client numbers, consistently demonstrate the excellent performance and robustness of our proposed FedIn-NID framework. The framework excels in handling heterogeneous, large-scale IIoT networks, providing significant improvements over baselines across all settings.

V. CONCLUSION AND FUTURE WORK

The advent of the next-generation Industrial IoT has intensified the demand for robust, scalable, and decentralized network security mechanisms. This paper presented a novel FedIn-NID framework to effectively tackle the issues arising from heterogeneous data distributions, large numbers of devices, and varying device availability in realistic IIoT environments. We introduced a novel multidimensional client selection strategy that integrates various factors to ensure efficient and balanced client participation. This strategy effectively mitigates the attack data heterogeneity among participating clients and accommodates the diverse availability of these clients. Additionally, we proposed FedEMA, a dynamic global model aggregation strategy to balance the global model and local models, achieving a robust and efficient integration of knowledge. Extensive experiments demonstrated the effectiveness of FedIn-NID, particularly its robust detection performance as data distribution heterogeneity increases and the number of devices grows, showcasing its superior overall performance compared to existing solutions. In summary, FedIn-NID offers a harmonious blend of scalability, adaptability, and robustness, positioning it as a cornerstone for future research and development aimed at safeguarding the increasingly complex and distributed world of next-generation IIoT.

To build upon this foundation, our future work will address both the practical deployment challenges and the algorithmic adaptability of the framework. It is important to note that our current investigation focused on fundamental algorithmic solutions within an idealized communication environment. Therefore, our first direction for future work is to evaluate and enhance the proposed method on standardized FL benchmark platforms. This will allow us to analyze its performance under realistic communication constraints and optimize its

system-level efficiency for practical, production-level deployment. Our second direction is to enhance the framework's adaptability to novel threats by integrating incremental learning capabilities. This will empower individual clients to locally learn emerging, previously unseen attack classes, with the federated learning framework then efficiently propagating this critical knowledge to the global model. These advancements will ensure the system can adapt more quickly to the ever-evolving threat landscape of real-world IIoT systems.

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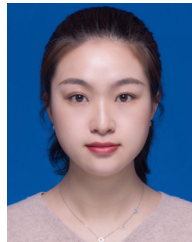
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